Leo Xu

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Technical Skills

Languages: Java, Python, R, C/C++, C#, Kotlin, JavaScript, SQL, Bash, TypeScript, HTML/CSS
Data/ML: NumPy, Pandas, Matplotlib, scikit-learn, TensorFlow, PyTorch, Apache Spark
Frameworks: React, Node.js/Express, Qt5, Unity, JUnit, Mockito
Tools & Technologies: Git, Docker, RStudio, VS Code/Visual Studio, Ansible, MongoDB, MySQL, Linux, Postman

Education

University of Waterloo — Waterloo, ON
Master of Data Science and Artificial Intelligence — *Sept. 2025 – Present*
- Relevant Courses: Machine Learning, Optimization, Databases, Concurrent Systems, Graph Theory

Carleton University — Ottawa, ON
Bachelor of Computer Science Honours — *Sept. 2019 – Apr. 2024*

Experience

Android Developer — Jan. 2023 – Aug. 2023
General Motors — *Markham, ON*
- Proactively modernized an unmaintained Android intent-testing tool by replacing manual compilation workflows, cutting emulator iteration time by ∼50% and significantly improving developer feedback loops
- Identified slow emulator feedback as a team-level bottleneck by comparing iteration latency across teams
- Improved consistency of intent-validation results across emulator runs, reducing false conclusions
- Implemented intent-driven UI state rendering for fast visual verification of emulator behavior
- Onboarded and guided new team members by documenting tooling and walking through debugging workflows

Data Scientist — May 2022 – Aug. 2022
Public Services and Procurement Canada — *Ottawa, ON*
- Migrated 200k+ procurement notices with a Python scraper (BeautifulSoup) replacing manual transfer
- Used Pandas to clean and validate procurement notice data by checking schema consistency, missing fields, and record counts, ensuring all public records were transferred correctly during the CanadaBuys migration
- Identified and documented 100+ defects during migration QA, enabling faster diagnosis by engineering teams
- Collaborated with developers and QA to prioritize high-impact issues and support migration readiness
- Supported the CanadaBuys platform launch through migration validation and post-release issue investigation

DevOps & Automation Engineer — Sept. 2021 – Dec. 2021
Royal Bank of Canada — *Toronto, ON*
- Built an ML-based anomaly detection workflow to monitor production API traffic, surfacing 50+ previously undetected usage and traffic anomalies during routine monitoring
- Routed anomaly signals to application support teams for faster investigation
- Tuned detection logic to reduce false positives and maintain alert usefulness for on-call engineers
- Automated alerting pipelines using Ansible to streamline incident response
- Extended monitoring coverage to legacy IMS databases, improving backend system visibility

Projects

Daymark – Intelligent Productivity & Job Engine | *React, TypeScript, Python* — Sept. 2026 – Jan. 2026
- Built an automated personal metrics dashboard that generates smart goal suggestions and trend indicators using rolling 7-day productivity averages
- Engineered a burnout prediction dashboard using activity pattern analysis to identify behavioral risk indicators
- Built a job-resume matching engine that utilizes skill extraction to calculate compatibility scores for applicants
- Integrated Google Calendar API for real-time data ingestion to automate scheduling and conflict detection

Used Car Price Predictor | *Python, Pandas, NumPy, scikit-learn, TensorFlow* — Sept. 2025 – Dec. 2025
- Built an end-to-end ML pipeline to predict used-car prices from structured vehicle and equipment data
- Achieved optimal RMSE and R^2 metrics using 5-fold cross-validation to ensure model robustness
- Conducted feature importance analysis to quantify the impact of brand and equipment on market valuation

Social Dynamics in Survival Scenarios | *Python* — Jan. 2024 – April. 2024
- Leveraged OpenAI Whisper and Pyannote-Audio to generate speaker-labeled transcripts from group dialogue
- Engineered NLP pipeline in Python to quantify sentiment and model interactions as weighted graphs
- Modeled evolving social ties as weighted graphs to quantify relationship valence and group cohesion
- Applied graph theory metrics to predict team performance based on communication patterns and leadership shifts